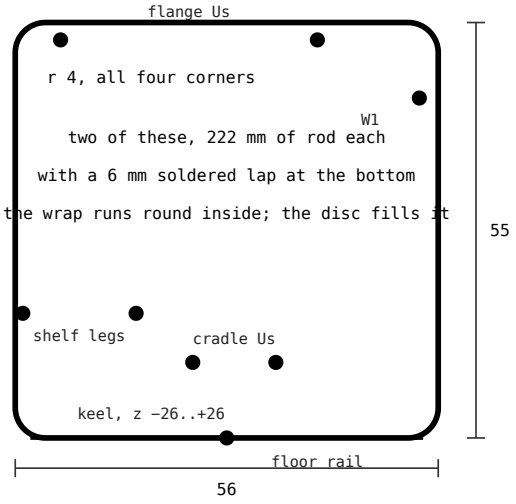
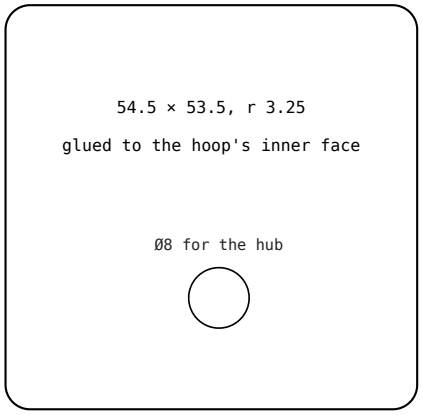
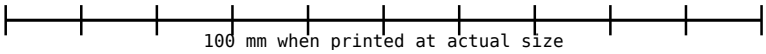
Mochi templates, 1:1 · Rev B, gear wheels · page 1 of 2 · hoop profile and side disc

Print at Actual size, not Fit. Check the bar. Bend each hoop of 1.5 mm brass rod on the outline (rod centreline), r 4 at the four corners, lap 6 mm at the bottom centre and solder. The dots are where the 1.0 mm rods between the two hoops meet this profile; the disc is the card that fills each hoop, glued to its inner face.







Piece	Qty	Rod	mm each	Where
hoop	2	1.5 mm	222	the outline, 56 × 55 r 4, lap-soldered at the bottom centre
floor rail	1	1.0 mm	56	hoop to hoop at y 5, z 0, under the floor skin
keel	1	1.0 mm	52	along z at x 0 under the floor, soldered across the floor rail; carries the two domes
cradle U	2	1.0 mm	76	y 15, z -4.5 and 6.5, legs 10 down to the hoops' bottoms: the motors sit on them
shelf	1	1.0 mm	204	a 54 × 15 rectangle at y 21.5 (z −27..−12) on four 16.5 legs to the hoops' bottoms: the holder sits on it
flange U	2	1.0 mm	61	y 57.7, z −22 and +12, legs 2.3 up to the hoops' tops: the lidar's flange rests on them
W1	1	1.0 mm	61	y 50, z 25.5, legs 2.5 forward to the hoops' fronts: the screen hangs on it
brim ring	1	1.0 mm	127	Ø40.5 ring on the top skin round the lidar body
domes	2	4 mm tube	6	a stub with a soldered dome, tail on the desk, nose 1 mm up
hub	2	5 over 4 mm tube	9	telescoped and soldered: the 4 mm tube's bore takes the 3 mm D-shaft, the 5 mm tube fills the gear's bore; boss inboard, flush with the gear's outer face

Mochi templates, 1:1 · Rev B, gear wheels · page 2 of 2 · the card wrap

One strip of cardstock, 209 × 54.5 mm, running inside the hoops: back, top, front, floor, with a 3 mm arc between each pair (the shaded bands). Cut every opening but the lidar hole before spraying; cut the lidar hole last, to the flange you measure. The hatch stays a flap: cut three sides.

